Movable-tray clearing: one bowl at a time

01

Observe

Acquire a fresh RGB-D frame.

02

Measure

Locate the tray and all bowls.

03

Verify

Check the previous placement.

04

Choose

One source bowl + one free slot.

05

Pick & place

Run the existing motion sequence.

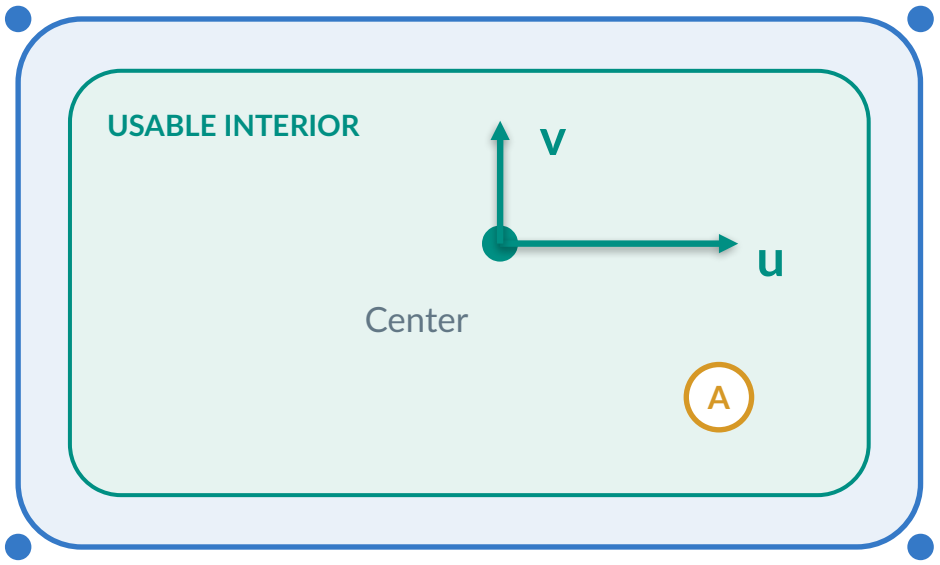
06

Return

Go back to the saved home pose.

Fresh scene each round • Same grasp and motion • Move the tray only between actions

YOLO locates the tray; depth gives its physical position



Bowls: YOLO mask + depth → footprint

Tray: YOLO mask + depth → center and heading

Segment the tray directly

A separate YOLO model supplies the tray outline.

Recover XY and heading

Same-frame depth + camera-to-base transform.

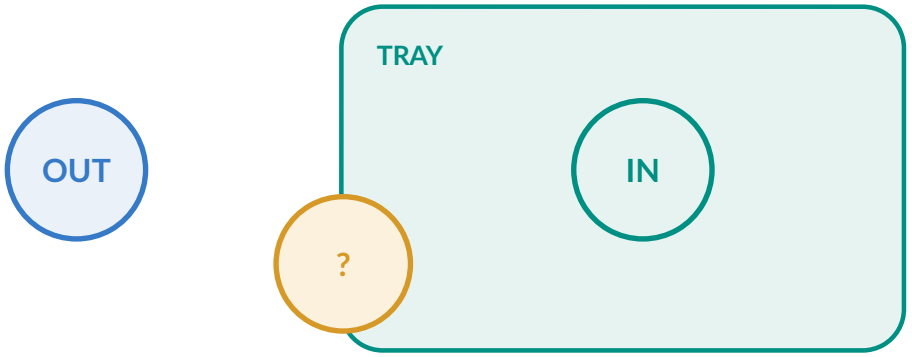
Move the slots with the tray

$p_{slot} = c_{tray} + R(\theta) \cdot [u, v]$

Dedicated tray segmentation weights required • outline and coordinate axes are schematic

Pick outside the tray. Place only in verified free space.

1 SELECT THE BOWL



Outside + in source region → pick candidate

Fully inside → occupancy only; never re-pick

Boundary / unclear depth → observe again

2 SELECT THE SLOT



Edge clearance ≥ bowl radius + margin

Bowl spacing > both radii + clearance

Keep used slots reserved if detections drop out.

Keep the motion sequence; update the destination XY

Same rim contacts and finger direction. Same seven motion segments.



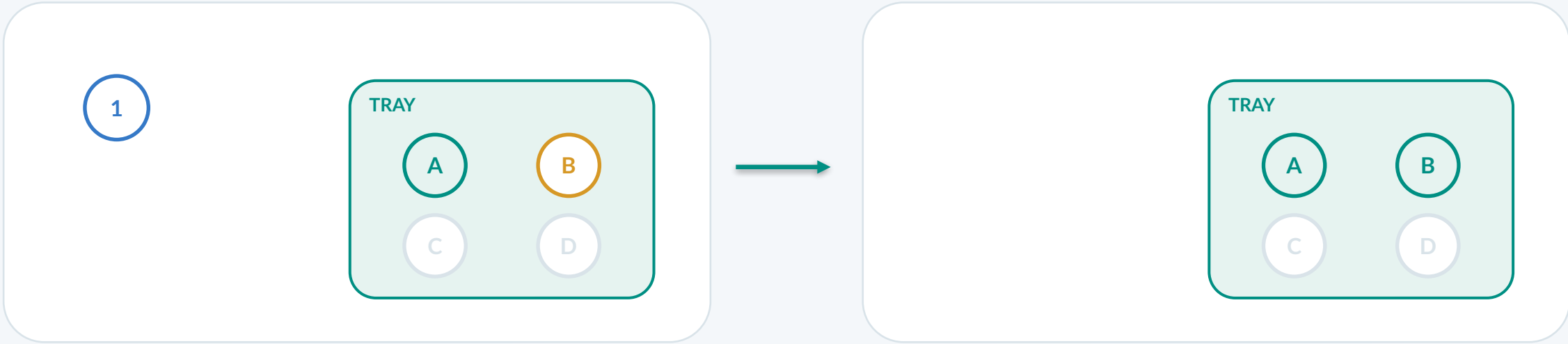
tool release XY = selected slot XY + fixed offset

Same Z • orientation • descent • home

Confirm a changed scene before starting the next pick

BEFORE THE ACTION

FRESH OBSERVATION AFTER RETURN



1

Source bowl is gone

2

New occupancy at the chosen slot

3

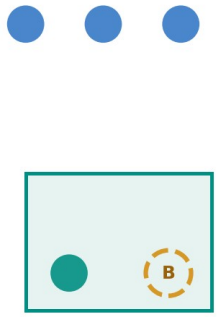
Previous tray occupancy remains

An action acknowledgment reserves the slot. A fresh observation confirms placement.

Three placements, five observations

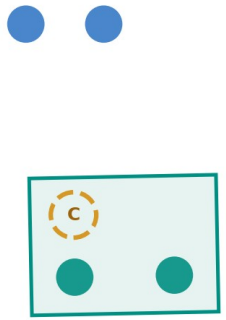
Blue: pending bowls • Teal: inside tray • Amber ring: selected destination

OBSERVATION 1



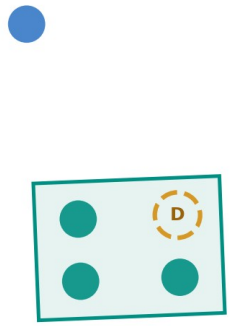
Pick → B

OBSERVATION 2



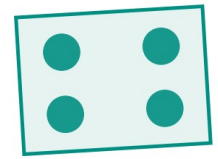
Confirm 1
Pick → C

OBSERVATION 3



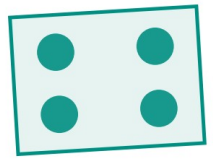
Confirm 2
Pick → D

OBSERVATION 4



Confirm 3
Check empty

OBSERVATION 5



Confirm empty
Complete

3 bowls confirmed / Slots B, C, D used / Existing slot A preserved

Four checks protect detection and placement decisions

01 Model contract

A bowl model has no trained tray class.

Check local weights, segmentation output and tray class.

02 Complete view

A partial outline can suggest a misleading free area.

Check depth, measured size, corners and the full view.

03 Stable slot direction

A rectangular mask can flip its reported axis by 180°.

Keep the closest direction; test occupied-slot continuity.

04 Bowl clearance

A large bowl needs more space than a small one.

Use the actual footprint; retain previously occupied slots.

Software checks passed; hardware trials are next

888

automated tests passed

881 regression + 7 ROS tests

3 packages built

14 **ROS transport**

Fresh messages and task identities

32 **Launch contracts**

Topics, namespaces and parameters

2 **Exit cleanup**

No owned workers left running

16 **MoveIt integration**

Planning and simulated motion

Before real execution

Tray model → calibrate → preview → test one bowl

Thin, flat tray • stationary during motion • no new 3D collision scene